

Deep Reinforcement Learning Assisted Trajectory Design for UAV Communication Systems Against Malicious Jamming

Jilan Jiao[†], Chenxi Liu[†], Jianhua He[‡], and Mugen Peng[†]

[†]State Key Laboratory of Networking and Switching Technology,

Beijing University of Posts and Telecommunications, Beijing, 100876, China

[‡]Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences, 100094, China

Email: {jjl_0620, chenxi.liu, pmg}@bupt.edu.cn, hejianhua@csu.ac.cn

Abstract—In this paper, we propose a novel trajectory design for the unmanned aerial vehicle (UAV) communication systems, where a single UAV moves along with multiple mobile terrestrial users to provide communication services, in the presence of malicious jamming. In our proposed design, we take the users' mobility and the jammer's adaptive jamming power into consideration and model the UAV's trajectory as a Markov decision process. Then, with the goal of optimizing the overall average throughput in our system, we design a twin-delayed deep deterministic policy gradient-based algorithm to judiciously optimize the trajectory of the UAV in a continuous manner, while simultaneously satisfying practical constraints on both the UAV's and users' mobility. We compare the performance of our proposed design with various baselines and show how it can improve the average throughput performance in the presence of malicious jamming and converge at a faster rate.

Index Terms—Unmanned aerial vehicle communications, trajectory design, mobile users, anti-jamming, deep reinforcement learning.

I. INTRODUCTION

The advancement of unmanned aerial vehicles (UAVs) has enabled a wide range of applications across military, civilian, and commercial sectors [1]. Compared to terrestrial communication systems, the UAV communication systems can further augment the coverage and capacity performance, due to the line-of-sight links and controllable mobility [2]. Against this background, UAV communications have attracted significant research attention [3].

In UAV communications, the trajectory is a key design problem, since it offers a new degree of freedom for performance improvements [4]–[7]. For example, in [4], the UAV's trajectory, the UAV's transmit power, and the user association were jointly optimized to maximize the minimum transmission rate among all the users. In [5] and [6], the UAV's trajectory was planned to minimize the time required to finish the task. In [7], the trajectories of multiple UAVs and the power control were jointly optimized to minimize the overall uplink transmit power of the internet-of-things (IoTs) devices.

The aforementioned works [4]–[7] mainly studied the UAV's trajectory design in the scenarios without malicious jamming. However, UAV communication systems are highly

vulnerable to jamming attacks, especially in military applications. When exposed to malicious jamming, the performances of the designed trajectories obtained from the existing works will be significantly degraded, making the trajectory optimization that is robust to the scenarios with malicious jamming an important research topic. In fact, considerable research efforts have been dedicated to examining the trajectory design problem in the presence of malicious jamming (e.g., [8]–[11]). Specifically, in [8], an uplink throughput maximization problem was investigated. Considering the impacts of malicious jamming, the three-dimensional (3D) trajectories of the UAVs and the transmission power of terrestrial users were jointly optimized. In [9], the scenarios of multiple jammers with incomplete location information were considered, and the UAV trajectories were designed to enhance the minimum uplink transmission rate of the users. In [10], the overall throughput of the UAV uplink communication system was maximized via the trajectory planning the UAVs. In [11], a worst-case signal-to-interference-plus-noise ratio (SINR) maximization problem was studied, in which the trajectory of the UAV within a limited flight duration was optimized. We note that these researchers commonly assumed that the terrestrial users remain stationary and the jamming power is fixed. This assumption, however, limits their applicability to the scenarios of mobile users and the jammer of adaptive power.

To solve these problems, we propose a novel trajectory design for the UAV communication systems against malicious jamming. In our considered system, a single UAV moves along with multiple terrestrial users to provide communication services. Compared to the existing works, our contributions are summarized as follows: 1) We investigate a practical UAV communication system, in which the users' mobility and the jammer's adaptive jamming power are taken into account. To optimize the average throughput of our system, we formulate a UAV's trajectory optimization problem, subject to practical constraints on the UAV's and the users' mobility. 2) We design a twin-delayed deep deterministic policy gradient (TD3)-based algorithm to solve the formulated problem. 3) We demonstrate how our proposed algorithm can enhance the average throughput performance with a faster convergence

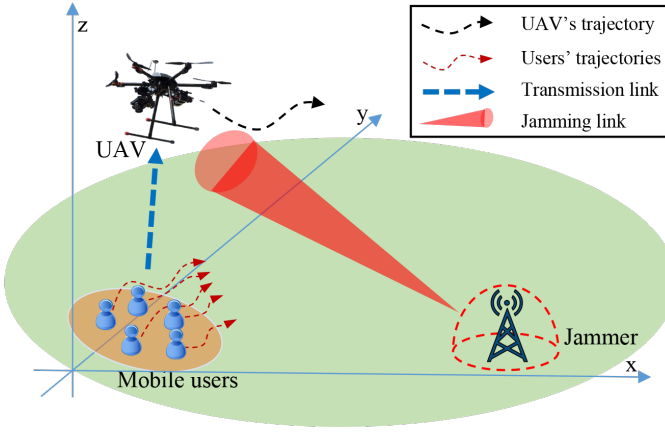


Fig. 1. Illustration of our considered UAV systems with malicious jamming.

rate, compared to various baselines.

II. SYSTEM MODEL AND PROBLEM FORMULATION

As depicted in Fig. 1, an uplink UAV-enabled network is considered, in which multiple mobile users, indexed by $\mathcal{M} = \{1, 2, \dots, M\}$, transmit to a UAV. The transmissions between the users and the UAV are interfered by a malicious jammer. In order to provide the communication service to the users, we consider that the UAV moves along with them. We denote the overall service time as T , and divide it into N equal time slots, indexed by $\mathcal{N} = \{1, 2, \dots, N\}$. We further divide N into K sub time slots, each with a δ_t duration, i.e. $N = K\delta_t$. As such, we have $T = NK\delta_t$ (as in [12]). We assume that users' positions remain constant within each sub time slot, while the position of the UAV remains constant in a given time slot. We note this assumption is practical in scenarios where the users' speed is smaller than the UAV's speed. The UAV's 3D coordinates at the n -th slot is $\mathbf{q}_n = [x_n, y_n, H]^T$, and the users' positions at the (n, k) sub-time slot is $\mathbf{q}_{m,(n,k)} = [x_{m,(n,k)}, y_{m,(n,k)}, 0]^T$. We assume that the jammer's coordinate, i.e., $\mathbf{q}_j = [x_j, y_j, 0]^T$, is fixed during the service time T and available to the UAV-enabled network¹.

A. Users' Mobility Model

In this subsection, we present the adopted users' mobility model. Specifically, we consider that each user moves subject to a Gaussian Markov random movement model [13]. Note that this model is suitable for the cases where the users' positions are time-dependent. Hence, it can avoid sudden changes in position. According to the Gaussian Markov ran-

dom movement model, at sub-time slot (n, k) , the mobile user m 's velocity and moving direction can be expressed as

$$v_{m,(n,k)} = l_1 v_{m,(n,k)} + (1 - l_1) \bar{v}_m + \sigma_{v_m} \sqrt{1 - l_1^2} g_{v_m,(n,k-1)}, \quad (1)$$

$$\theta_{m,(n,k)} = l_2 \theta_{m,(n,k)} + (1 - l_2) \bar{\theta}_m + \sigma_{\theta_m} \sqrt{1 - l_2^2} g_{\theta_m,(n,k-1)}, \quad (2)$$

where $\bar{v}_m$ and $\bar{\theta}_m$ represent user m 's average velocity and moving direction, respectively. σ_{v_m} and σ_{θ_m} denote the standard deviation of v_m and θ_m , respectively. g_{v_m} and g_{θ_m} are random variables with mean zero and variance one. l_1 and l_2 are used to describe the level of randomness in the users' mobility.

Based on (1) and (2), the two-dimensional (2D) coordinate of mobile user m at sub time slot (n, k) is expressed as

$$x_{m,(n,k)} = x_{m,(n,k-1)} + v_{m,(n,k-1)} \cos(\theta_{m,(n,k-1)}), \quad (3)$$

$$y_{m,(n,k)} = y_{m,(n,k-1)} + v_{m,(n,k-1)} \sin(\theta_{m,(n,k-1)}). \quad (4)$$

In addition, we adopt a probabilistic line-of-sight (LoS) channel model. As such, the LoS probability for the link connecting the UAV and user m (jammer j) can be represented as

$$\eta^{\text{LoS}}(\theta_{\xi,(n,k)}) = \frac{1}{1 + A_2 \exp(-A_1(\theta_{\xi,(n,k)} - A_2))}, \quad (5)$$

where A_1 and A_2 are the parameters determined by the communication environment [14], $\theta_{\xi,(n,k)}$ represents the angle of elevation between user m -th user (jammer j) and the UAV at sub time slot (n, k) .

According to (5), we express the path loss between user m (jammer j) and the UAV at the sub time slot (n, k) as

$$PL_{\xi,(n,k)} = \eta^{\text{LoS}}(\theta_{\xi,(n,k)}) PL_{\xi,(n,k)}^{\text{LoS}} + \eta^{\text{NLoS}}(\theta_{\xi,(n,k)}) PL_{\xi,(n,k)}^{\text{NLoS}}, \quad \xi \in \{m, j\}, \quad (6)$$

where $\eta^{\text{NLoS}}(\theta_{\xi,(n,k)}) = 1 - \eta^{\text{LoS}}(\theta_{\xi,(n,k)})$, $PL_{\xi,(n,k)}^{\text{LoS}}$ and $PL_{\xi,(n,k)}^{\text{NLoS}}$ represent the pathloss of LoS link and NLoS link, respectively, given by

$$PL_{\xi,(n,k)}^{\text{LoS}} = 20 \log(4\pi f_c d_{\xi,(n,k)} / c) + \eta^{\text{LoS}}, \quad (7)$$

$$PL_{\xi,(n,k)}^{\text{NLoS}} = 20 \log(4\pi f_c d_{\xi,(n,k)} / c) + \eta^{\text{NLoS}}. \quad (8)$$

In (7) and (8), f_c and c denote the carrier frequency and the velocity of light, respectively. η^{LoS} and η^{NLoS} are the mean of the additional losses for the LoS and NLoS conditions, respectively. $d_{\xi,(n,k)} = \|\mathbf{q}_n - \mathbf{q}_{\xi,(n,k)}\|$ represents the distance between the UAV and user m . The channel gain between the UAV to user m (jammer j) at sub time slot (n, k) is represented as $h_{\xi,(n,k)} = 10^{-PL_{\xi,(n,k)}/10}$.

As such, the received SINR of user m at sub time slot (n, k) is given by

$$\gamma_{m,(n,k)} = \frac{p_{m,(n,k)} h_{m,(n,k)}}{p_{j,n} h_{j,n} + \sigma^2}, \quad (9)$$

¹The location of the jammer can be estimated through a measurement campaign of its jamming signals.

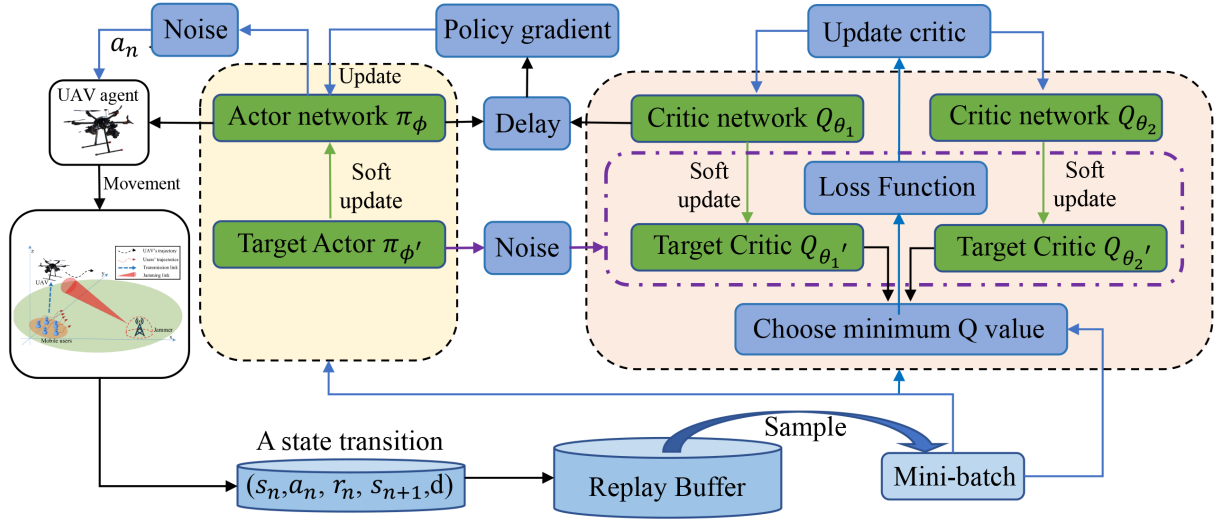


Fig. 2. Framework of the proposed TD3-based Algorithm.

where $p_{\xi,(n,k)}$ represents the user m 's (jammer j) transmit power, and σ^2 represents the power of the additive white Gaussian noise.

According to (9), we can compute the transmission rate for user m at the time slot n by

$$\mathbf{R}_{m,n} = B \sum_{k=1}^L \log_2 \left(1 + \frac{p_{m,(n,k)} h_{m,(n,k)}}{p_{j,n} h_{j,n} + \sigma^2} \right). \quad (10)$$

B. Problem Formulation

We aim to optimize the throughput of all the mobile users through the design of the UAV's trajectory $\mathcal{Q} = \{\mathbf{q}_n\}_{n=1}^N$, subject to practical mobility constraints. Mathematically, this problem can be expressed as

$$\max_{\{\mathcal{Q}\}} \frac{1}{N} \sum_{n=1}^N \sum_{m=1}^M \mathbf{R}_{m,n} \quad (11a)$$

$$\text{s.t. } C1: \|\mathbf{q}_n - \mathbf{q}_{n-1}\| \leq V_{\max} L \delta_t, \quad (11b)$$

$$C2: 0 \leq x_n \leq x_{\max}, 0 \leq y_n \leq y_{\max}, \quad (11c)$$

$$C3: 0 \leq x_{m,(n,k)} \leq x_{\max}, 0 \leq y_{m,(n,k)} \leq y_{\max}, \quad (11d)$$

$$C4: \theta_{n-1} \leq \theta_{\max}, n \geq 2, n \in \mathcal{N}, \quad (11e)$$

$$C5: \mathbf{R}_{m,n} \geq \mathbf{R}_{\min}, n \in \mathcal{N} \quad (11f)$$

$$C6: 0 \leq p_j \leq p_{j,\max} \quad (11g)$$

where constraint (C1) is to ensure that the maximum flying distance during a time slot is within a certain range. Meanwhile, constraint (C2) guarantees that the UAV is within the designated region, and (C3) prohibits the users from moving beyond the prescribed area. In addition, (C4) imposes a limitation on the maximum turning angle at $\mathbf{q}_n$. Considering the fairness of all users, constraint (C5) ensures that each user at the time slot (n, k) achieves the minimum necessary transmission rate.

Finally, (C6) ensures that the jammer's power is below the maximum level.

We would like to note that the problem in (11) is difficult to address since it is a continuous optimization problem over the UAV's trajectory. Traditionally, this problem is solved by transforming into multiple waypoints optimization problems and only a sub-optimal solution can be obtained. To address this problem, we introduce a deep reinforcement learning (DRL) approach to intelligently design the UAV's trajectory considering both the users' movement and the jammer's transmission power. The proposed approach will be detailed in Section III.

III. PROPOSED TD3-BASED TRAJECTORY DESIGN

In this section, a TD3-based solution is proposed to address the problem in (11). We note that the TD3-based method has shown to be effective in dealing with reinforcement learning (RL) [15] tasks with continuous action space, therefore is particularly suitable for solving our considered problem in (11).

In order to apply the TD3-based method, we first characterize the UAV's trajectory (i.e., the waypoints across N time slots) as a Markov decision process (MDP). For this purpose, we use a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \beta \rangle$ to represent this MDP, where the state space is denoted by $\mathcal{S}$, the action space is denoted by $\mathcal{A}$, the reward function is denoted by $\mathcal{R}$, the state transition probability is denoted by $\mathcal{P}$ and the discount factor is denoted by β . In each time slot, considering the state $s \in \mathcal{S}$, an action of the UAV is chosen $a \in \mathcal{A}$ based on its policy, and gets a reward r .

State space, action space, and reward function are defined as follows:

- **State Space $\mathcal{S}$:** At the time slot n , the system state $s_n \in \mathcal{S}$ can be represented as $s_n = [\mathbf{q}_{m,(n,k)}; \mathbf{q}_n; p_j; L_{sh}]$, where L_{sh} is a punishment when the UAV's coordinate violates constraint C2 in (11).

- **Action Space $\mathcal{A}$:** The action space $\mathcal{A}$ includes all potential actions of the UAV. Specifically, at the time slot n , the action a_n can be defined as $a_n = [\theta_n, v_n]$. We note that both θ_n and v_n are continuous variables. Furthermore, the problem in (11) is a continuous optimization problem.
- **Reward r_n :** At the time slot n , the UAV's reward r_n from action a_n is defined as

$$r_n = \sum_{m=1}^M \mathbf{R}_{m,n} + L_{sh}, \quad (12)$$

where $\sum_{m=1}^M \mathbf{R}_{m,n}$ is the throughput of all mobile users.

Furthermore, we define the policy as the mapping from $\mathcal{S}$ to $\mathcal{A}$, i.e., $\pi : \mathcal{S} \rightarrow \mathcal{A}$, consisting of the state transition probability, i.e., $\pi(a|s) = Pr(a_n = a|s_n = s)$. We also define the experience as $e_n = (s_n, a_n, r_n, s_{n+1})$, which is saved in an experience replay memory $\mathcal{E}$.

Next, we present our proposed TD3-based algorithm that solves the trajectory of the UAV optimization problem in (11). Fig. 2 illustrates the framework of our proposed algorithm, which consists of a training network and a target network. More specifically, each network can be further divided into an actor network and two critic networks. In addition, we note that in Fig. 2 the experience relay buffer $\mathcal{E}$ is used to store the collected experience e_n during the training and update the parameters of the networks.

We now focus on how the TD3-based algorithm works. Specifically, after the initialization, for time slot n , the UAV chooses an appropriate action a_n on the basis of the initial policy of the actor network, i.e., π_ϕ , and the current state s_n . Then, the UAV obtains the corresponding reward r_n based on (12) and updates the state as s_{n+1} . In addition, the experience e_n is stored in the experience relay memory $\mathcal{E}$.

Then, the Q -value approximator $Q_\theta(s_n, a_n)$ with parameter $\theta \in \{\theta_1, \theta_2\}$ is updated by minimizing the loss function $F_{loss}(\theta)$, given by

$$F_{loss}(\theta) = \frac{1}{N_a} \sum_{n=1}^N (y_n - Q_\theta(s_n, a_n))^2. \quad (13)$$

where N_a represents the size of the minibatch, the value of the target is represented by

$$y_n = r_n + \beta Q_{\theta'}(s_{n+1}, a_{n+1}), a_{n+1} \sim \pi_{\phi'}(s_{n+1}). \quad (14)$$

In (14), the subsequent state is represented by s_{n+1} , $\pi_{\phi'}$ represents that an action a_{n+1} is chosen from the target actor network, and $Q_{\theta'}$ represents the target critic networks, $\theta' \in \{\theta'_1, \theta'_2\}$.

The policy of the actor network can be adjusted through the use of the deterministic policy gradient algorithm [16], which is represented by

$$\nabla_\phi J(\phi) = \frac{1}{N_a} \sum_{n=1}^N [\nabla_{a_n} Q_\theta(s_n, a_n)|_{a_n=\pi_\phi(s_n)} \nabla_\phi \pi_\phi(s_n)]. \quad (15)$$

Algorithm 1 The proposed algorithm

```

1: Initialize  $Q_{\theta_1}, Q_{\theta_2}, \pi_\phi$  with random parameters  $\theta_1, \theta_2, \phi, \theta'_1, \theta'_2$ , and  $\phi', \mathcal{E}$ 
2: for  $episode = 1 \rightarrow M$  do
3:   Receive  $s_0$ , let  $t = 0$  and  $d = 0$ 
4:   repeat
5:     Choose an action  $a_t$ , and examine  $r_t$  and  $s_{t+1}$ 
6:     if the UAV traverses the limitation then
7:        $\psi_t = \psi_t - L_{sh}$ , and revise  $r_t, s_{t+1}$ 
8:     end if
9:     if the UAV finishes the user service quest, i.e.,  $\sum_{k=1}^K \sum_{t=1}^N c_{t,k} = T$  then
10:       the episode comes to an end, i.e.,  $d = 1$ 
11:     end if
12:     Save the  $(s_t, a_t, r_t, s_{t+1}, d)$  in  $\mathcal{E}$ 
13:     if  $|\mathcal{E}| > 3000$  then
14:       From  $\mathcal{E}$ , sample a mini-batch  $B$  transition set.
15:       Update  $\tilde{a}$  and  $\tilde{y}$ 
16:       Update the parameters  $\theta_i$  of the critic networks
17:       Update the policy of action  $\phi$ 
18:       Update the networks of the target:
19:        $\theta'_i \leftarrow \kappa \theta_i + (1 - \kappa) \theta'_i$ 
20:        $\phi' \leftarrow \kappa \phi + (1 - \kappa) \phi'$ 
21:     end if
22:     Update  $n \leftarrow n + 1$ 
23:     until  $\sum_{l=1}^L \sum_{n=1}^N c_{n,l} = T$ 
24:   end for
```

The target critic networks and the target actor are updated based on the critic networks and the actor network, respectively, and are expressed as

$$\theta' \leftarrow \kappa \theta + (1 - \kappa) \theta', \quad (16)$$

$$\phi' \leftarrow \kappa \phi + (1 - \kappa) \phi', \quad (17)$$

where κ is the rate of decay for the actor network and the critic networks. Our proposed algorithm is presented in Algorithm 1.

We note that our TD3-based algorithm integrates three distinct techniques to mitigate the overestimation bias problem, i.e., clipped double Q -learning, target policy smoothing regularization, and delayed policy updates. We detail each technique as follows:

- **Clipped double Q -learning.** In the TD3-based framework, Q_{θ_i} ($i = 1, 2$) are two Q -functions of it. The critic networks and target critic networks contain two Q networks. The Q target network with a lower Q -value is employed to evaluate the target Q -value in each update. As such, the target update procedure is given by

$$y_n = r_n + \beta \min_{i=1,2} \{Q_{\theta'_i}(s_{n+1}, \tilde{a})\}, \quad (18)$$

where r_n represents the specified reward and $\tilde{a}$ represents the output from the target policy.

- **Target policy smoothing regularization.** We use regularization in our proposed TD3-based algorithm to mitigate

TABLE I
SIMULATION PARAMETERS LIST

parameters	Value
Episodes	3000
Exploration noise	$\epsilon \sim \mathcal{N}(0, 0.36)$
Decay rate	$\sigma = 0.999$
Experience relay buffer capacity	$\mathcal{E} = 1 \times 10^5$
Soft-update rate of the target network	$\kappa = 0.005$
The factor of discount	$\beta = 0.99$
Learning rate of the networks	$l_r = 0.0001$
Time step	$N_{max} = 200$

the variability in the target outcome. Specifically, we add the clipped random noise into the target action, and perform smoother value estimation by using mini-batches with size B , given by

$$\tilde{a} = \pi_{\phi'}(s_{n+1}) + \tau, \tau \sim \text{clip}(\mathcal{N}(0, \tilde{\omega}), -b, b). \quad (19)$$

Then, the updated principle about the TD3 algorithm is given by [17]

$$\theta_i \leftarrow \arg \min_{\theta_i} \frac{1}{B} \sum (y_n - Q_{\theta_i}(s_n, a_n))^2, \quad (20)$$

$$\nabla_{\phi} J(\phi) = \frac{1}{B} \sum \nabla_{a_n} Q_{\theta_1}(s_n, a_n)|_{a_n=\pi_{\phi}(s_n)} \nabla_{\phi} \pi_{\phi}(s_n) \quad (21)$$

- *Delayed policy updates.* We note that, in the proposed TD3 framework, the target network and the target actor network update less often compared to the critic networks and the target critic networks. This updated strategy can effectively reduce the cumulative error of the actor network, making the actor network more stable and thus contributing to a faster convergence rate of algorithm 1.

IV. NUMERICAL RESULTS

In this section, the numerical results are presented to verify the efficacy of the proposed algorithm. We consider a simulated area of size $400 \text{ m} \times 400 \text{ m}$, the number of terrestrial users $M = 5$. The UAV's maximum velocity is set as $V_{max} = 40 \text{ m/s}$ and the users' average speed $\bar{v}$ is 4 m/s with $\sigma_v = 2$. The initial 3D coordinate of the UAV is $q_0 = [0, 0, H_0]^T$ with $H_0 = 150 \text{ m}$. The jammer's 3D coordinate is $q_j = [1000 \text{ m}, 0, 0]^T$, the entire flight duration T is set as 360 s , and the time period of a sub time slot is 1 s , the time period of a time slot is 5 s . In addition, the noise level is $\sigma^2 = -114 \text{ dBm}$, the maximum transmit power is $P_{m,max} = 10 \text{ W}$. We consider that the jamming power is 80 W . Furthermore, the additional path loss for the LoS link is 0.1 dB , while the NLoS link incurs an additional path loss of 21 dB [14]. The carrier frequency is set as 2 GHz . The simulation parameters are given in Table I.

In Fig. 3, we verify the convergence of the proposed algorithm. In this figure, we depict the cumulative reward achieved by our proposed algorithm versus the number of episodes. For comparisons, we depict the convergence performance of two baseline methods, namely, the DDPG-based algorithm and

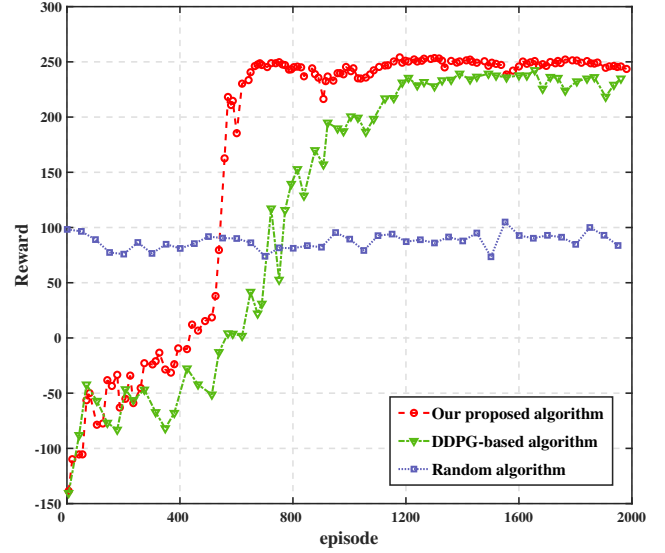


Fig. 3. Convergence performance of different schemes.

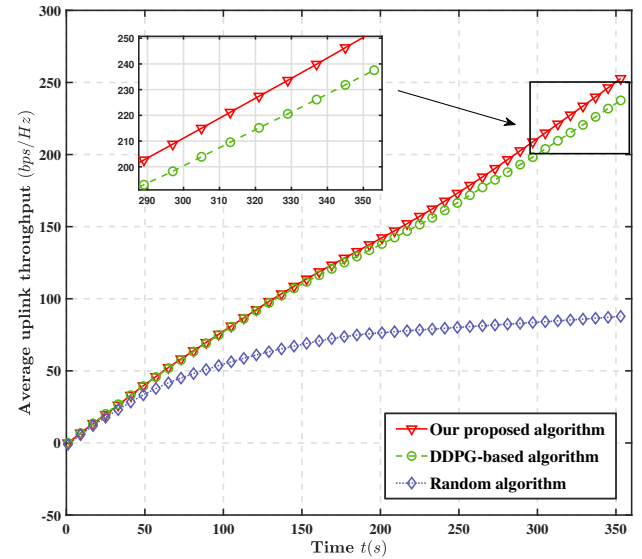


Fig. 4. Average uplink throughput of different schemes for $P_j = 100 \text{ W}$.

the random algorithm. Specifically, the DDPG-based algorithm generates the UAV's trajectory according to [18], while in the random algorithm, the direction of the UAV is set randomly between 0 and 2π , and the velocity of the UAV is set randomly between 0 and 40 m/s . We find that our proposed algorithm converges faster than the baselines. Moreover, we note that the proposed algorithm gets a substantially higher cumulative reward compared to the baseline methods, demonstrating the effectiveness and advantages of our proposed algorithm.

In Fig. 4, we simulate the average throughput of the UAV across time when using different algorithms. In this figure, we

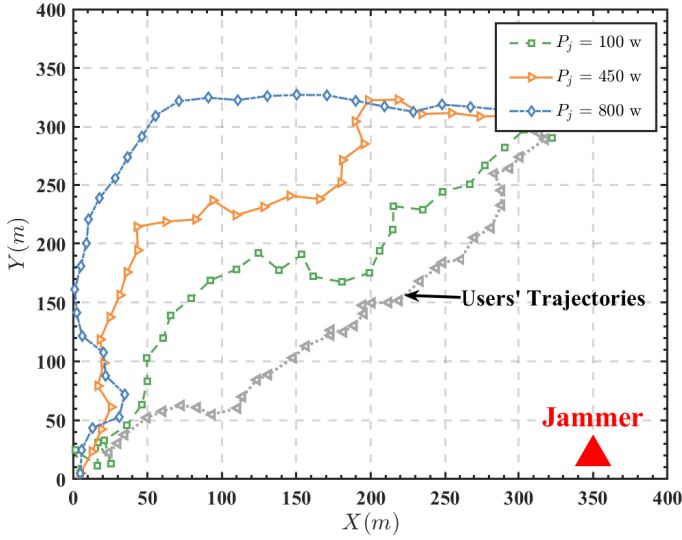


Fig. 5. The UAV's trajectory for different jamming power.

consider that the jamming power $P_j = 100$ W. We observe that the average throughput achieved by our proposed algorithm increases with time t . We then see that, compared to the baselines, our proposed algorithm can achieve the highest average throughput, which once again verifies the effectiveness of our proposed algorithm.

Finally, in Fig. 5, we examine the impacts of malicious jamming on the UAV's trajectory. Specifically, we depict the trajectories generated by our proposed for varying values of P_j . We also plot the users' trajectory (denoted by gray triangles) for reference. We first see that, for all the values of P_j , the UAV moves as the users move. We also see that, as P_j increases, the UAV tends to fly further away from the jammer, indicating how our algorithm can be robust to the adaptive jamming power.

V. CONCLUSION

In this paper, we proposed a TD3-based trajectory design for UAV communication systems, taking the impacts of mobile users and malicious jamming into account. Specifically, we first formulated an average throughput maximization problem, subject to the constraints of user mobility and the UAV's kinematics. Then, we leveraged the benefits of the TD3-based algorithm and optimized the UAV trajectory of continuous action. The numerical results show that our proposed algorithm is more robust to malicious jammers in terms of the average throughput performance, compared to the various baselines.

ACKNOWLEDGMENT

This work was supported in part by National Key R&D Program of China (No. 2021YFB2900200), in part by the National Natural Science Foundation of China (No. 62001047), in part by the Young Elite Scientist Sponsorship Program by China Institute of Communications, and in part by the Youth Innovation Promotion Association CAS.

- [1] Y. Zeng, R. Zhang, and T. J. Lim, "Wireless communications with unmanned aerial vehicles: Opportunities and challenges," *IEEE Commun. mag.*, vol. 54, no. 5, pp. 36–42, May. 2016.
- [2] Y. Wu, J. Li, J. Liu, M. Sheng, and C. Zhao, "Enabling efficient caching in high mobility UAV communications network under limited backhaul," *China Commun.*, vol. 19, no. 10, pp. 207–219, Oct. 2022.
- [3] L. Zhang, H. Zhao, S. Hou, Z. Zhao, H. Xu, X. Wu, Q. Wu, and R. Zhang, "A survey on 5G millimeter wave communications for UAV-assisted wireless networks," *IEEE Access*, vol. 7, pp. 117460–117504, Jul. 2019.
- [4] Q. Wu, Y. Zeng, and R. Zhang, "Joint trajectory and communication design for multi-UAV enabled wireless networks," *IEEE Trans. Wireless Commun.*, vol. 17, no. 3, pp. 2109–2121, May. 2018.
- [5] Y. Zeng, X. Xu, and R. Zhang, "Trajectory design for completion time minimization in UAV-enabled multicasting," *IEEE Trans. Wireless Commun.*, vol. 17, no. 4, pp. 2233–2246, Apr. 2018.
- [6] J. Zhang, Y. Zeng, and R. Zhang, "UAV-enabled radio access network: Multi-mode communication and trajectory design," *IEEE Trans. Signal Process.*, vol. 66, no. 20, pp. 5269–5284, Oct. 2018.
- [7] M. Mozaffari, W. Saad, M. Bennis, and M. Debbah, "Mobile unmanned aerial vehicles for energy-efficient internet of things communications," *IEEE Trans. Wireless Commun.*, vol. 16, no. 11, pp. 7574–7589, Sep. 2017.
- [8] Y. Gao, Y. Wu, Z. Cui, H. Chen, and W. Yang, "Robust design for turning and climbing angle-constrained UAV communication under malicious jamming," *IEEE Commun. Lett.*, vol. 25, no. 2, pp. 584–588, Feb. 2021.
- [9] Y. Wu, W. Fan, W. Yang, X. Sun, and X. Guan, "Robust trajectory and communication design for multi-UAV enabled wireless networks in the presence of jammers," *IEEE Access*, vol. 8, pp. 2893–2905, Dec. 2020.
- [10] H. Wang, J. Chen, G. Ding, and J. Sun, "Trajectory planning in UAV communication with jamming," in *Proc. IEEE WCSP 2018*, Hangzhou, China, Oct. 2018.
- [11] Z. Lingyun, Z. Xiaotong, G. Xin, S. Enbin, Z. Xin, and S. Qingjiang, "Robust trajectory planning for UAV communication systems in the presence of jammers," in *Proc. Chinese J. Aeronaut.*, 2022, Haikou, China, May. 2009.
- [12] X. Zhang, C. Liu, and M. Peng, "Three-dimensional trajectory designs for unmanned aerial vehicle-enabled communications with kinematic constraints," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10910–10922, Oct. 2022.
- [13] D. Li, J. Zhou, J. Wang, and P. Yu, "Linking generation rate based on Gauss-Markov mobility model for mobile ad hoc networks," in *Proc. IEEE NSWCTC 2009*, Wuhan, China, May. 2009.
- [14] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal lap altitude for maximum coverage," *IEEE Wireless Commun. Lett.*, vol. 3, no. 6, pp. 569–572, Dec. 2014.
- [15] R. S. Sutton and A. G. Barto, "Reinforcement learning: An introduction," *Robotica*, vol. 17, no. 2, pp. 229–235, Dec. 1999.
- [16] D. Silver, G. Lever, N. Heess, T. Degris, D. Wierstra, and M. Riedmiller, "Deterministic policy gradient algorithms," in *Proc. ICML 2014*, Guangzhou, China, Aug. 2014.
- [17] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra, "Continuous control with deep reinforcement learning," *arXiv preprint arXiv:1509.02971*, Jul. 2015.
- [18] O. Bouhamed, H. Ghazzai, H. Besbes, and Y. Massoud, "Autonomous UAV navigation: A DDPG-based deep reinforcement learning approach," in *Proc. IEEE ISCAS 2020*, Seville, Spain, Sep. 2020.